Sam (Peset) Tan

Senior Software Engineer

Newark, California | pesettan@gmail.com | pesettan.github.io
linkedin.com/in/sam-peset-tan-84932683

Summary

Senior software engineer building multitenant healthcare applications and the shared infrastructure behind them. Experience spans C#/.NET services, Kafka event delivery, audit systems, PostgreSQL, ClickHouse, and Kubernetes on AWS, with Angular and TypeScript experience across application interfaces and service integration.

Technical Skills

Backend: C#, .NET, ASP.NET Core, TypeScript, Node.js, Python, Go

Distributed systems and data: Microservices, Kafka, transactional outbox, multitenancy, PostgreSQL, ClickHouse, Redis

Cloud and delivery: AWS, Kubernetes, Helm, Codefresh CI/CD

Frontend: Angular, Nx, reusable components

Quality and reliability: Unit and integration testing, concurrency and regression testing, performance benchmarking, operational diagnosis

Professional Experience

Senior Software Engineer | Omnicell | 2017-Present

Multitenant healthcare platform for medication inventory, hospital restocking, identity and authorization, and site configuration.

- Implemented reusable ASP.NET Core auditing and entity-change capture in shared backend infrastructure, recording actor, tenant, trace, and before/after values. Isolated concurrent request contexts and preserved existing package contracts while limiting service integration boilerplate.
- Optimized audit middleware through cached configuration and streamlined claim processing. In repeated in-process Release/.NET 10 benchmarks, the authenticated single-role scenario executed 32.6% faster with 28.2% fewer allocations than the prior implementation; validated request isolation and compatibility with regression tests.
- Implemented TypeScript/Node.js audit ingestion from tenant-specific Kafka topics into ClickHouse, retaining the existing Elasticsearch path. Added payload validation, deterministic event IDs, safe handling of large offsets, and offset commits after successful persistence; supported evolving entity-change schemas and tenant-specific connection lifecycles.
- Contributed to shared .NET transactional-outbox infrastructure using PostgreSQL and Kafka, persisting events with business changes and supporting leases, retries, and ordering by topic and message key. Optimized claim queries for uneven backlogs; synthetic local tests across 100 isolated tenants and three relays reconciled 299,100 committed events with no observed loss, duplication, misrouting, malformed records, or ordering errors. Retained at-least-once delivery semantics.

Omnicell Continued

- Diagnosed cross-service audit delivery gaps caused by a mismatch between the queue written and the queue polled. Added an audit-aware reader for fresh and older pending events, preserving stored payloads, keys, headers, and existing business-event behavior; added concurrency and compatibility regression coverage.
- Implemented Angular/Nx audit-search behavior and contributed reusable frontend components. Preserved row selection across pagination, limited page-level selection to visible rows, and reset selection on filter or sort changes. Added identity-service auditing for role changes and login events, including readable role names and privacy metadata.
- Diagnosed ClickHouse audit-query costs and unreadable transaction-report data. Developed guarded recovery runbooks that preserved readable values and required complete evidence before staging or cutover; validated the expanded tooling with 83 local tests. Troubleshot CI/CD, package compatibility, Kubernetes/Helm deployments, and Kafka lag and offsets.
- Served as the inaugural developer for a hospital restocking cloud application; helped migrate legacy desktop functionality into three backend services and two Angular applications on Kubernetes. Gathered initial requirements and contributed to the technical roadmap and architecture.

Web Developer and Data Analyst | ATAC | 2008-2017

- Developed C# tools to parse large aviation datasets, prepare Excel outputs, and integrate with proprietary Windows applications for analysts and airspace controllers.
- Cleaned and presented aviation data; prototyped Angular applications with Node.js backends to visualize flight, noise, weather, and historical trends.

Education

M.S. Software Engineering | Penn State

B.S. Electrical Engineering | San Jose State